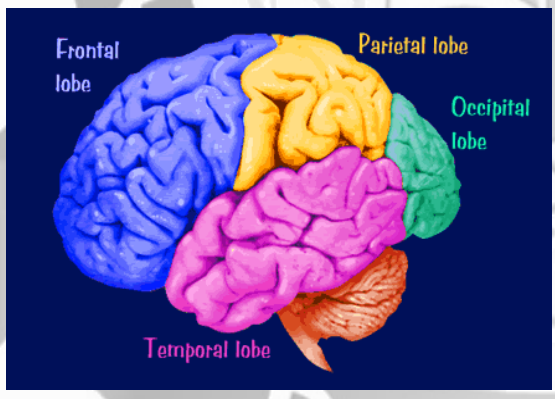
Microscopic Anantomy of Cerebral Cortex

Learning Objectives:

- ❑ At the end of the lecture, the students will be able to know:
- ❑ Review of gross anatomy of cerebral hemisphere.
- ❑ Know the types of cerebral cortices.
- ❑ Understand different types of cells of cerebral cortex.
- ❑ Discuss the layers of different cerebral cortices .

Cerebral Hemisphere:

- Forebrain consists of cerebral hemisphere.
- The cerebral hemisphere consists of neurons, neuroglial cells and fibers.
- Cells and most of neuroglial cells are present in superficial part of cerebral cortex, which appears gray on cross section, called as cortex .



- The fibers along with neuroglial cells are present in deeper part of cerebral hemisphere and fibers are myelinated that's why deeper part of cerebral hemisphere appear white on cross section

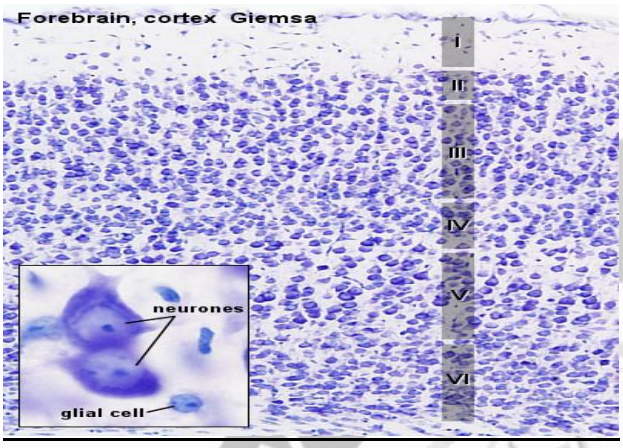
Types of Cerebral Cortex:

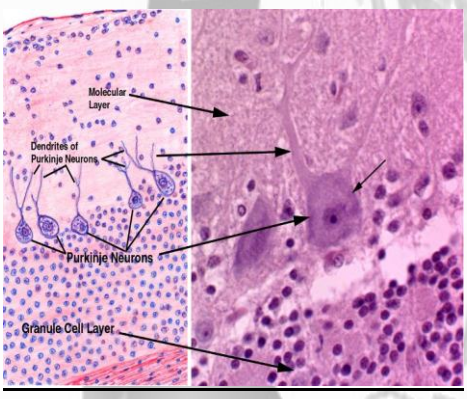
- Neocortex

- **Newest in evolution**
- **About 90% of total**
- **6 layers, most complex**
- **Paleocortex**
- **Associated with olfactory system, the parahippocampal gyrus,**
- **uncus**
- **fewer than 6 layers (3 layers)**
- **Archicortex**
- **Hippocampal formation; limbic system**
- **fewer than 6 layers (3-4 layers), most primitive**
- **Mesocortex**
- **Cingulate gyrus**
- **Transitional between archicortex and neocortex**
- **Histology of Cerebral Cortex:**

Histology of Cerebral Cortex:

- **Granule (stellate) cells** are interneurons
- Short dendrites extending in all directions
- Short axon projecting to adjacent pyramidal cells
- Granule cells are especially numerous in sensory and association cortex





Layers of Cerebral Cortex:

- Neocortex has 6 layers designated I, II, III, IV, V, VI
- Pyramidal cells predominate in layers III and V
- Granule cells in layers II and IV

Cerebral Cortex(layers):

plexiform (molecular): dendrites and axons of cortical neurons making synapses; neuroglia and rare horizontal cells

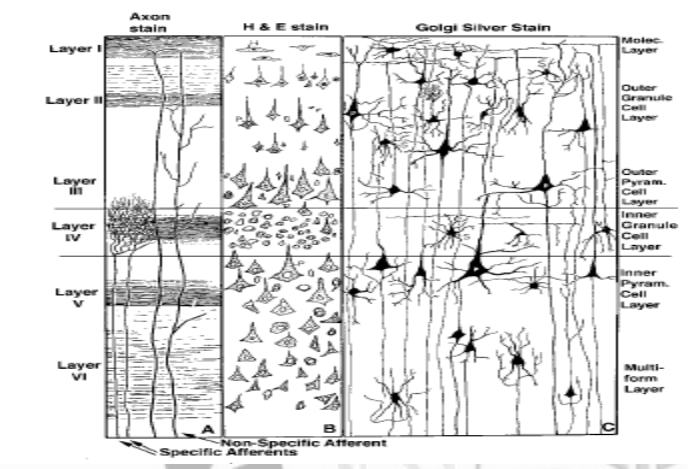
II: outer granular: dense population small pyramidal cells and stellate cells [small neurons]; various axons and dendritec connections

III: (outer) pyramidal cell: moderate size; increasing size deeper

IV: inner granular: densely packed stellate cells

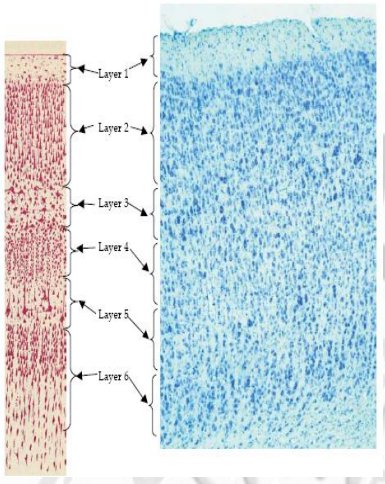
V: ganglionic or inner pyramidal: lg pyramidal cells (including Betz cells) and smaller numbers of stellate cells and small neurons .

VI: multiform (fusiform) cell: numerous small neurons, small pyramidal cells, stellate cells, especially superficially and fusiform cells in deeper part.



I: plexiform

II: outer granular cell



III: (outer) pyramidal cell

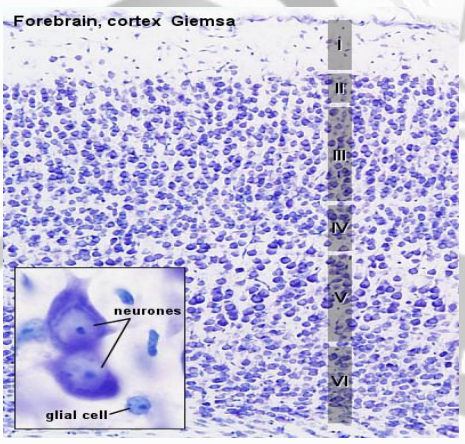
IV: inner granular

V: ganglionic or inner pyramidal

VI: multiform (fusiform) cell

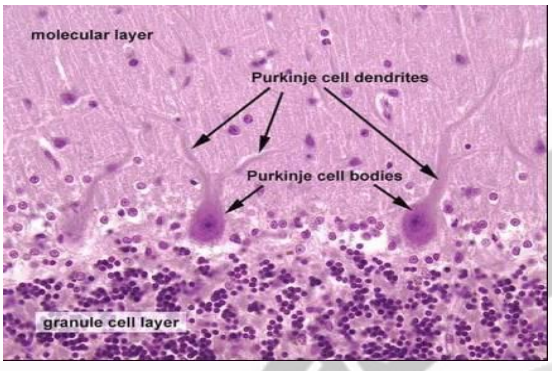
Layer I (the "molecular layer"):

- **Is the outermost layer.**
- **This layer contains relatively few nerve cell bodies.**
- **The odd name "molecular layer" derives from the fine texture of this layer, due to its composition largely of dendrites and fine axon terminals (and glia, of course).**

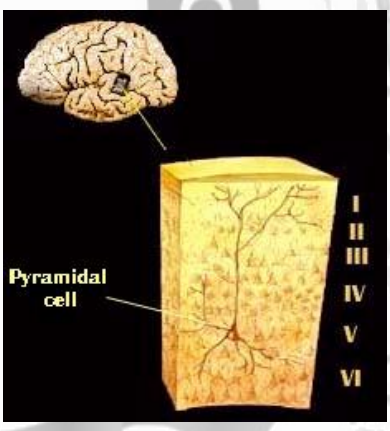


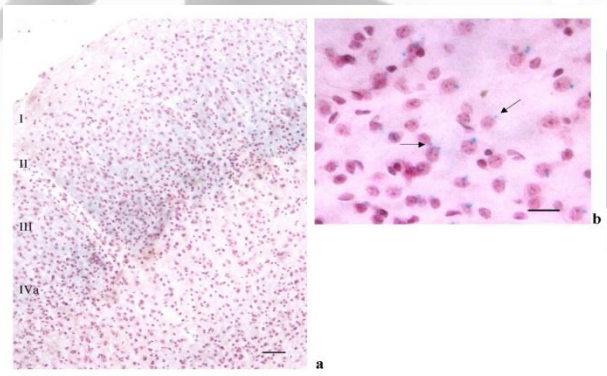
Layer II (the "outer granular layer"):

- **typically contains many very small cells (granule cells).**



- **Layer III (the "outer pyramidal layer") contains cell bodies of small pyramidal cells.**
- **Axons from these cells typically project to the upper layers of neighboring cortical regions.**

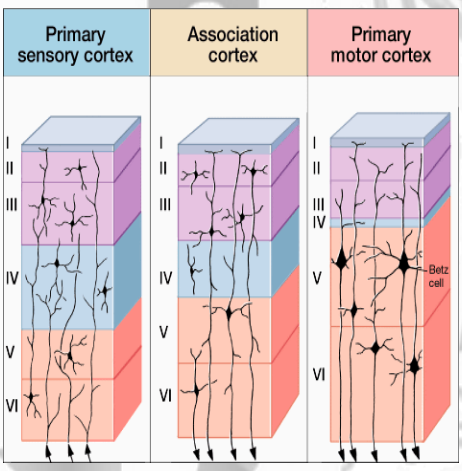


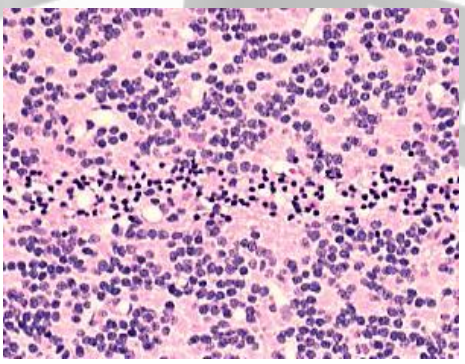




Layer IV (inner granular layer):

- **contains axonal ramifications of afferent fibers, such as sensory axons from the thalamus.**

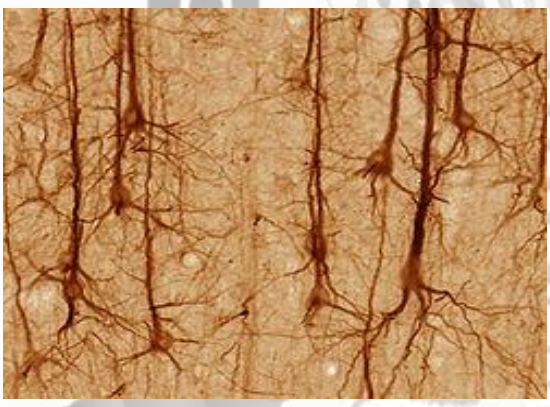




Layer V (the "inner pyramidal layer"):

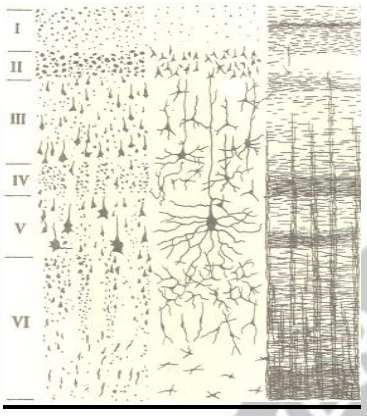
- **contains cell bodies of large pyramidal cells.**
- **Axons from these cells typically project to more distant cortical regions, to other parts of the brain, or to lower centers (such as spinal motor neurons).**
- **The larger size of these pyramidal cells (compared the smaller cells of layer III) is associated with the greater length of their axons.**

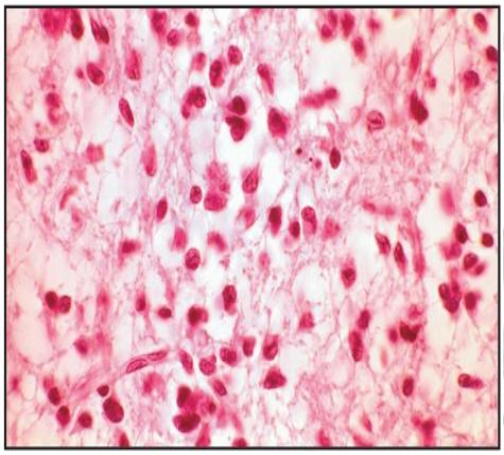




Layer VI (layer of pleiomorphic cells):

Layer VI (layer of pleiomorphic cells)





THANK YOU ALL!!!